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Master Thesis

The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

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by

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Abstract

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Contents

Summary	i
Nomenclature	iii
1 Introduction	1
2 Theory	3
2.1 The atmosphere under greenhouse gas forcing	3
2.2 Hadley circulation strength under greenhouse gas forcing	3
2.3 The angular momentum budget	3
2.3.1 Axisymmetric theory	3
2.3.2 Eddy driven mean circulation	5
2.3.3 konrad	5
2.4 The energy budget of the atmosphere	5
2.5 Model setup	7
2.5.1 ECHAM5	7
2.5.2 konrad	7
3 Methods	8
3.1 ECHAM5	8
3.2 konrad	8
3.3 Decomposition in forcing factors	8
4 Results	9
4.1 The atmospheric response	9
4.2 Predictions from axisymmetric scalings	10
4.3 Eddy driven Hadley cell predictions	13
4.4 The energetic framework	13
4.4.1 GMS and energy transport of the Hadley cell	13
4.4.2 Eddy energy transport	13
4.4.3 Forcing and feedback contributions	13
4.4.4 The forcing	13
5 Discussion	15
6 Conclusion	16
References	18

Nomenclature

Abbreviations

Abbreviation	Definition
USA	United States of America
EU	European Union

Variable Names

Symbol	Definition	Unit
α	greek alpha	[N]
β	greek beta	[J]

1

Introduction

Establishing of context: What is the HC?

The Hadley cell (HC) is the dominant meridional circulation in the tropics. Its ascending branch at the equator is collocated with inter-tropical convergence zone (ITCZ). The dry regions of subsidence in the subtropics are associated with the descending branch of the HC. In the annual mean there are two overturning cells in the tropics, which extend from 0 to 30 ° N in the Northern hemisphere and equivalently 0 to 30 ° S in the Southern hemisphere. Throughout the annual cycle there are strong variations in the HC's extent and strength: As the maximum of insolation moves poleward, so does the ITCZ and the HCs, resulting in a strong winter and a weaker summer HC. This seasonal migration is related to the global monsoons.

The HC is responsible for the poleward transport of angular momentum (and energy) and the equatorward transport of moisture, thereby reducing the meridional temperature gradients imposed by solar insolation. Therefore, the HC has strong implications for life on this planet. Its extent crucially modulates global precipitation patterns. The strength of the HC determines the moisture transport of moisture and energy. This is particularly given in the Sahel region. This semi-arid region receives most of its precipitation in the boreal summer when the HC and the ITCZ migrate northward. Thus, the rainfall in the Sahel is dynamically linked to the HC. Changes in the HC also affect the livelihood of the local population and economies which still very much depend on agricultural production. Paleoclimate simulations of the HC show large changes in HC strength and extend during the last glacial maximum and the mid holocene due to changes in orbital forcing (D'Agostino et al., 2017). This is also supported by evidence from tropical paleoclimate proxy records of precipitation changes (McGee et al., 2014).

These paleoclimate records combined with the large societal impact of precipitation patterns give a strong reason to investigate the projected HC changes of human driven greenhouse gas emissions in the current era.

CO₂ is very uniformly distributed in the atmosphere. The forcing that it imposes is not. The HC strength response depends on meridional forcing structure. This has been shown for quadrupling of CO₂ only in certain latitude belts (Kim et al., 2022). A forcing in mid latitudes weakens the Hadley circulation whereas a forcing at the equator leads to it strengthening. Thus, we want to investigate how CO₂ forcing in the vertical alters atmospheric circulation. We achieve this by quadrupling CO₂ only in the troposphere and only in the stratosphere using a pre-industrial control tropopause height. This approach is completely new and has not yet

been described by the literature. Previous studies emphasize the importance of drivers related to the troposphere to explain the HC weakening (Chemke & Polvani, 2021; Fan & Dommenget, 2024). (This study highlights vertical stability.) However, historical simulations show that also stratospheric forcing influences the general circulation. In particular stratospheric ozone depletion has led to a cooling of the Southern hemisphere lower-stratospheric polar cap. This has been identified as the main driver for the widening of the Hadley circulation over the second half of the twentieth century (Polvani et al., 2011).

We find that the HC response to tropospheric CO₂ is a strengthening and widening in both hemispheres. This opposes our expectations from the literature which sees tropospheric warming parameters as the main drivers for HC weakening with uniform CO₂ concentrations. Our results suggest that the weakening of the HC originates from stratospheric cooling caused by stratospheric CO₂. Our study disentangles the contributions of tropospheric and stratospheric CO₂ to the established HC weakening with uniform CO₂ distribution. To investigate the different responses of the HC strength we employ angular momentum and energy budget frameworks. The axisymmetric angular momentum framework to attribute HC changes to changes in tropopause height, meridional temperature, gradient, and vertical stability. Since this theory ignores the contribution of eddies to HC strength we also investigate changes in eddy divergence to explain the observed changes in the mean circulation. The energetic framework uses top-of-atmosphere (TOA) radiative fluxes and moist static energy to distinguish different contribution factors.

Significance

We contribute to the advancement of the field by revising the applicability of different HC frameworks and simplified models. Splitting up CO₂ forcing in two vertical layers puts existing theories to a test and gives the opportunity to rethink accepted truths in climate science. Overall is this a contribution to foundational research that advances our physical understanding on the drivers of general circulation changes. **Overview over the study** Is this really necessary?

2

Theory

2.1. The atmosphere under greenhouse gas forcing

tropospheric warming
lapse rate feedback
stratospheric cooling

2.2. Hadley circulation strength under greenhouse gas forcing

Which drivers have been identified for the weakening of the HC

2.3. The angular momentum budget

Why is there a circulation in the first place

2.3.1. Axisymmetric theory

The questions why there is a meridional circulation in the first place is answered by the axisymmetric theory (Held & Hou, 1980). This question is raised because one could imagine a world where Earth is just in radiative equilibrium everywhere. This means that every point on Earth emits just as much radiation as it receives. Thus, there would not be any meridional temperature/energy transport necessary. Given a temperature profile of the earth

$$\frac{\Theta_E}{\Theta_0} = 1 - \frac{1}{3}\Delta_h(3\sin^2\theta - 1) + \Delta_v\left(\frac{z}{H} - \frac{1}{2}\right) \quad (2.1)$$

obtained from radiative equilibrium conditions (see Figure 2.1), one can obtain a zonal wind via the thermal wind relationship. This not only produces zonal winds which are not observed in the atmosphere but also violates Hide's theorem. Instead we now assume an angular momentum conserving circulation

$$u_M = \Omega a \frac{\sin^2\theta}{\cos\theta} \quad (2.2)$$

and infer via the thermal wind relationship a temperature profile. Given that the energy is conserved in the upper branch one obtain a finite width of the HC and a measure for its strength.

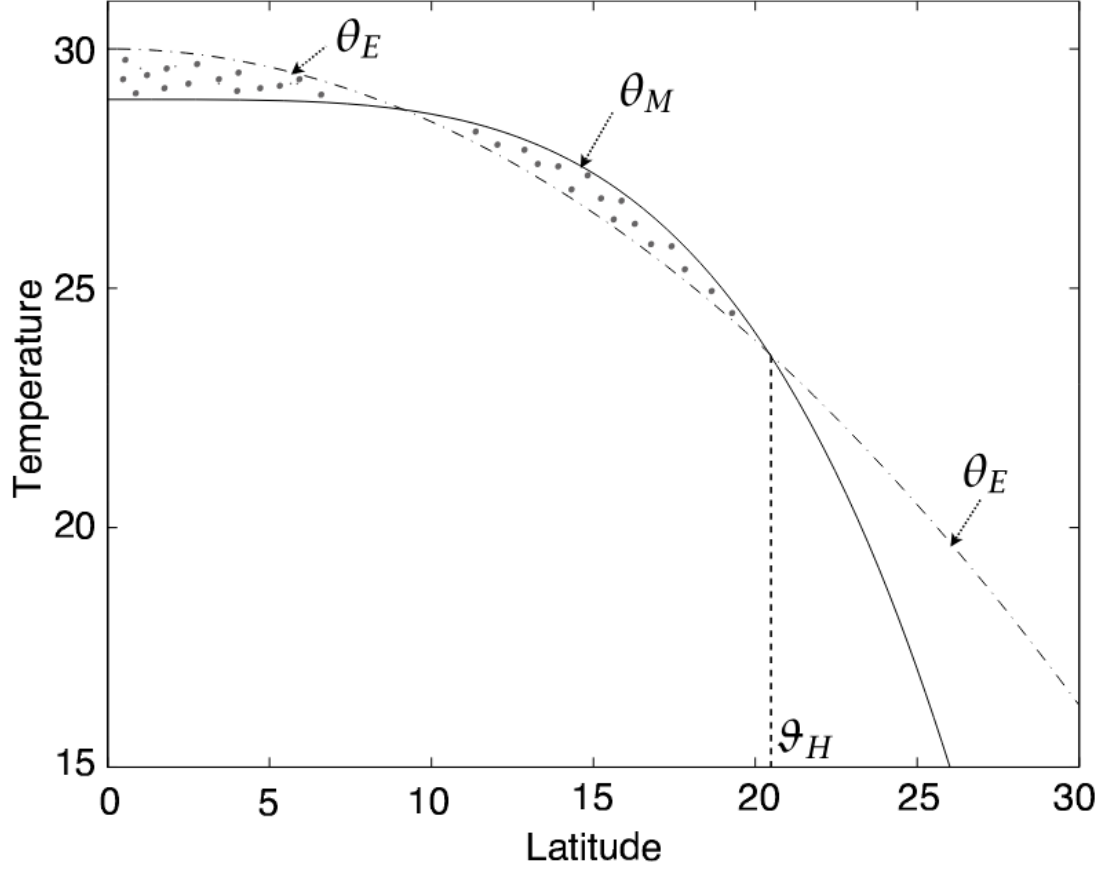


Figure 2.1: (Vallis, 2017)

$$\psi_{\max} \sim \frac{1}{\tau} \frac{g^{3/2} (H \Delta_h)^{5/2}}{\Delta_v a^2 \Omega^3} \quad (2.3)$$

$$\phi_H \sim \left(\frac{g H \Delta_h}{\Omega^2 a^2} \right)^{1/2} \quad (2.4)$$

$$\frac{\delta \psi_{\max}}{\psi_{\max}} = \frac{5}{2} \frac{\delta H}{H} + \frac{5}{2} \frac{\delta \Delta_h}{\Delta_h} - \frac{\delta \Delta_v}{\Delta_v} \quad (2.5)$$

$$\frac{\delta \phi_H}{\phi_H} = \frac{1}{2} \frac{\delta H}{H} + \frac{1}{2} \frac{\delta \Delta_h}{\Delta_h} \quad (2.6)$$

2.3.2. Eddy driven mean circulation

Eddies can drive a circulation (Ferrel cell) or as in the case of the HC significantly enhance it

$$f\bar{v}(1 - Ro) = \frac{1}{a \cos^2 \varphi} \frac{\partial(\cos^2 \varphi \overline{u'v'})}{\partial \varphi} + \frac{\partial \overline{u'\omega'}}{\partial p} \quad (2.7)$$

$$\psi_{\max} = \frac{2\pi a \cos \varphi}{g} \int_{p_{\max}}^0 \bar{v} dp' \quad (2.8)$$

$$= \frac{2\pi a \cos \varphi}{fg} \int_{p_{\max}}^0 \left(\frac{1}{a \cos^2 \varphi} \frac{\partial(\cos^2 \varphi \overline{u'v'})}{\partial \varphi} + \frac{\partial \overline{u'\omega'}}{\partial p} \right) dp' \quad (2.9)$$

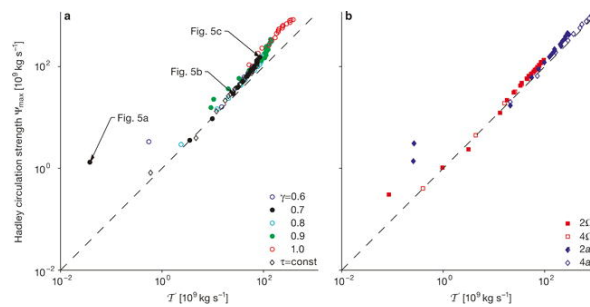


Figure 2.2: (Walker & Schneider, 2006)

2.3.3. konrad

2.4. The energy budget of the atmosphere

The energy budget analysis provides diagnostic perspective on the HC. It cannot explain why the Earth is not in radiative (convective) equilibrium. Using TOA radiative fluxes, one can infer that the short-wave radiative flux at the equator is much larger than the emitted outgoing long wave radiation. The opposite is true for the poles. This implies an energy transport by atmosphere and ocean. The atmospheric component can be further split up into a mean and an eddy component.

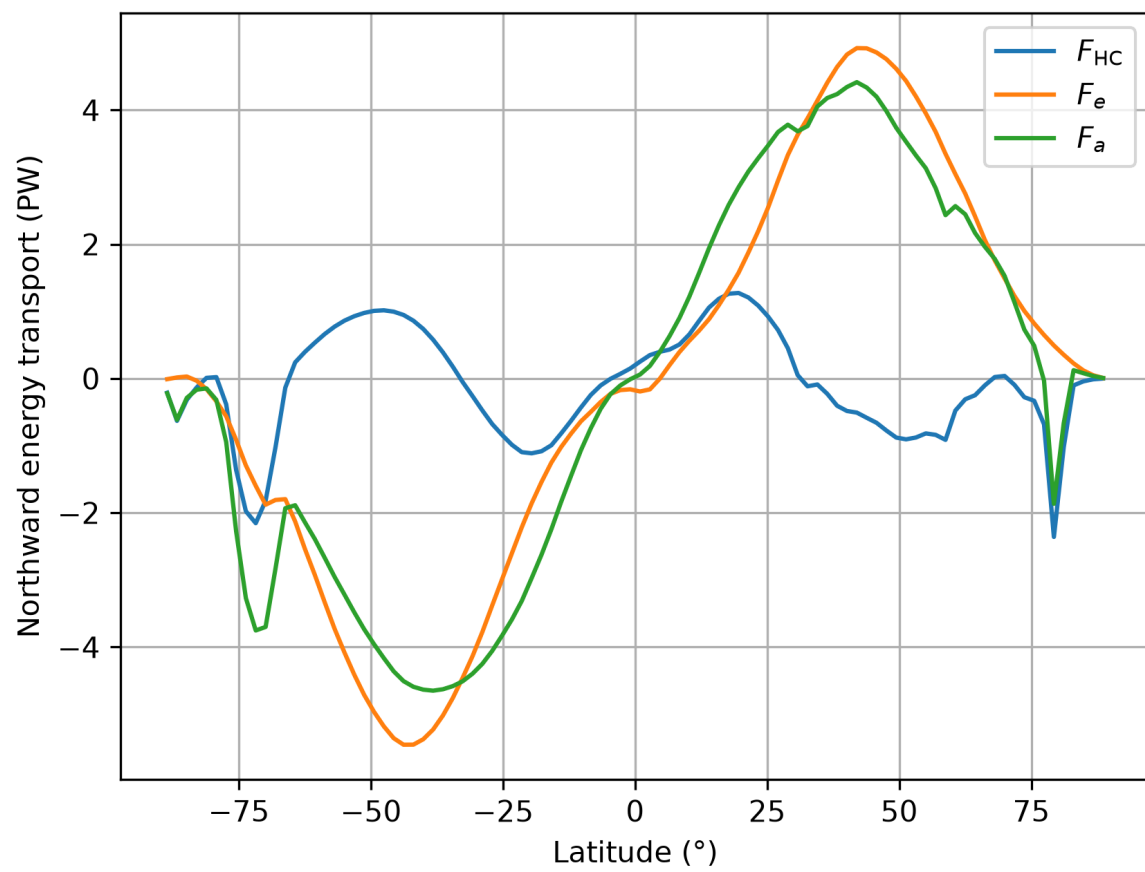


Figure 2.3

2.5. Model setup

2.5.1. ECHAM5

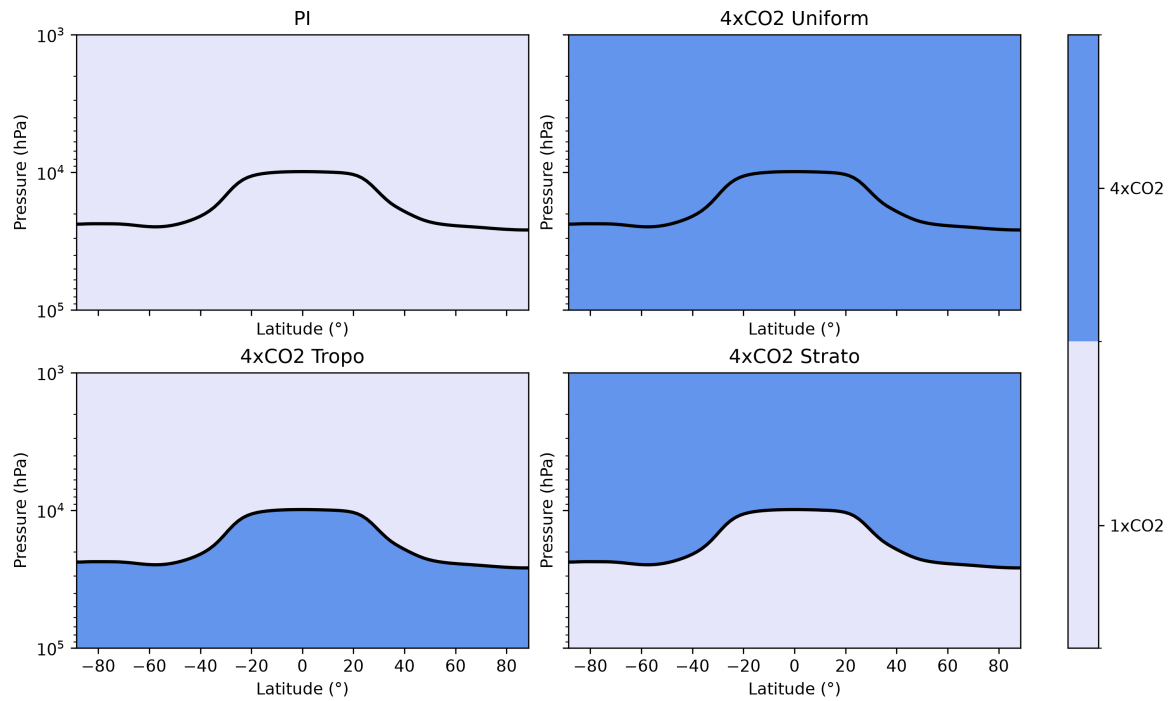


Figure 2.4

2.5.2. konrad

3

Methods

- 3.1. ECHAM5
- 3.2. konrad
- 3.3. Decomposition in forcing factors

4

Results

4.1. The atmospheric response

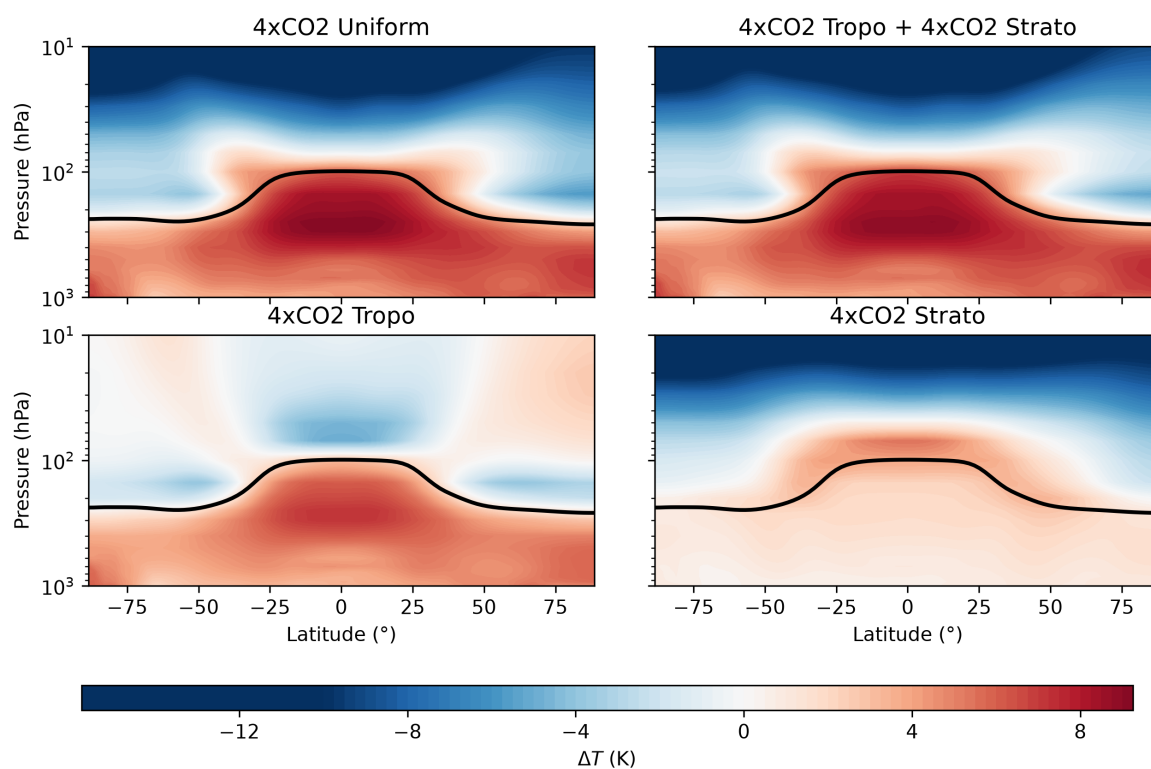


Figure 4.1

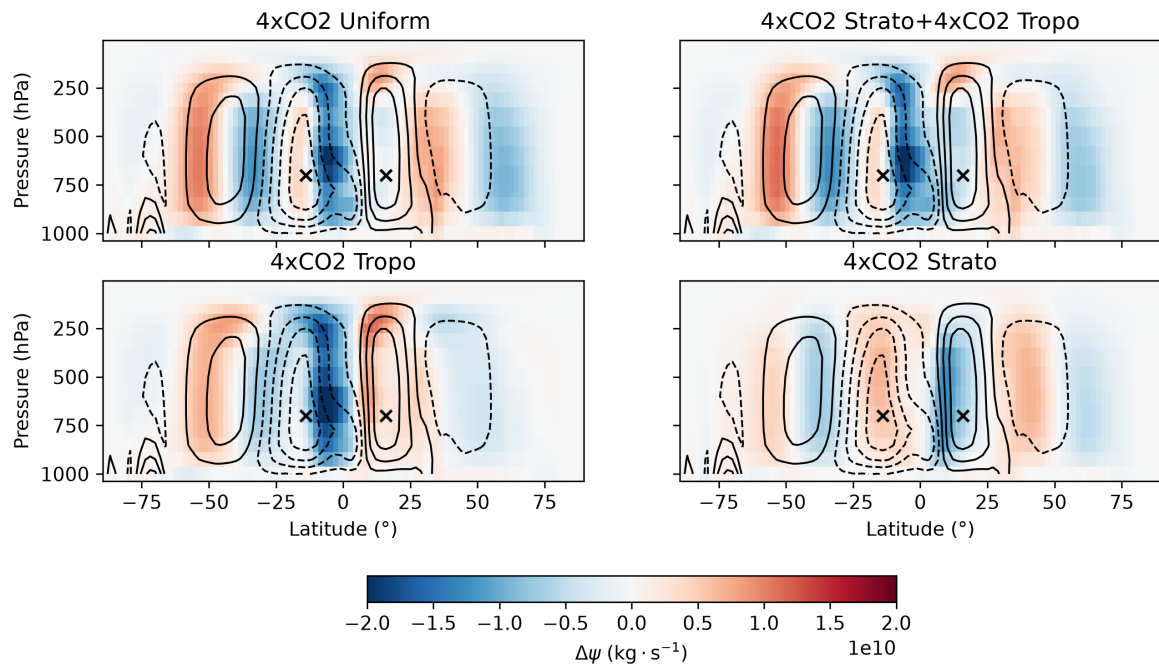


Figure 4.2

4.2. Predictions from axisymmetric scalings

(Seo, Frierson, & Son, 2014)

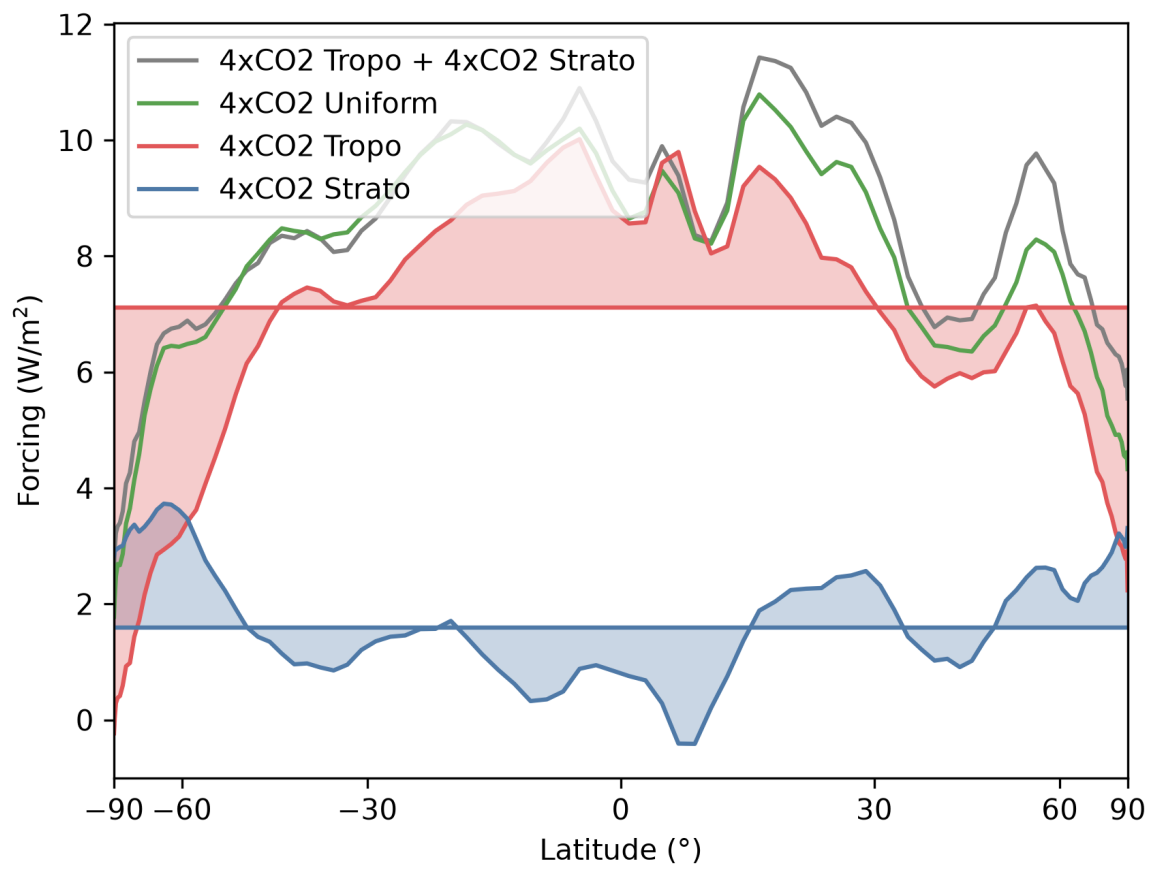


Figure 4.3

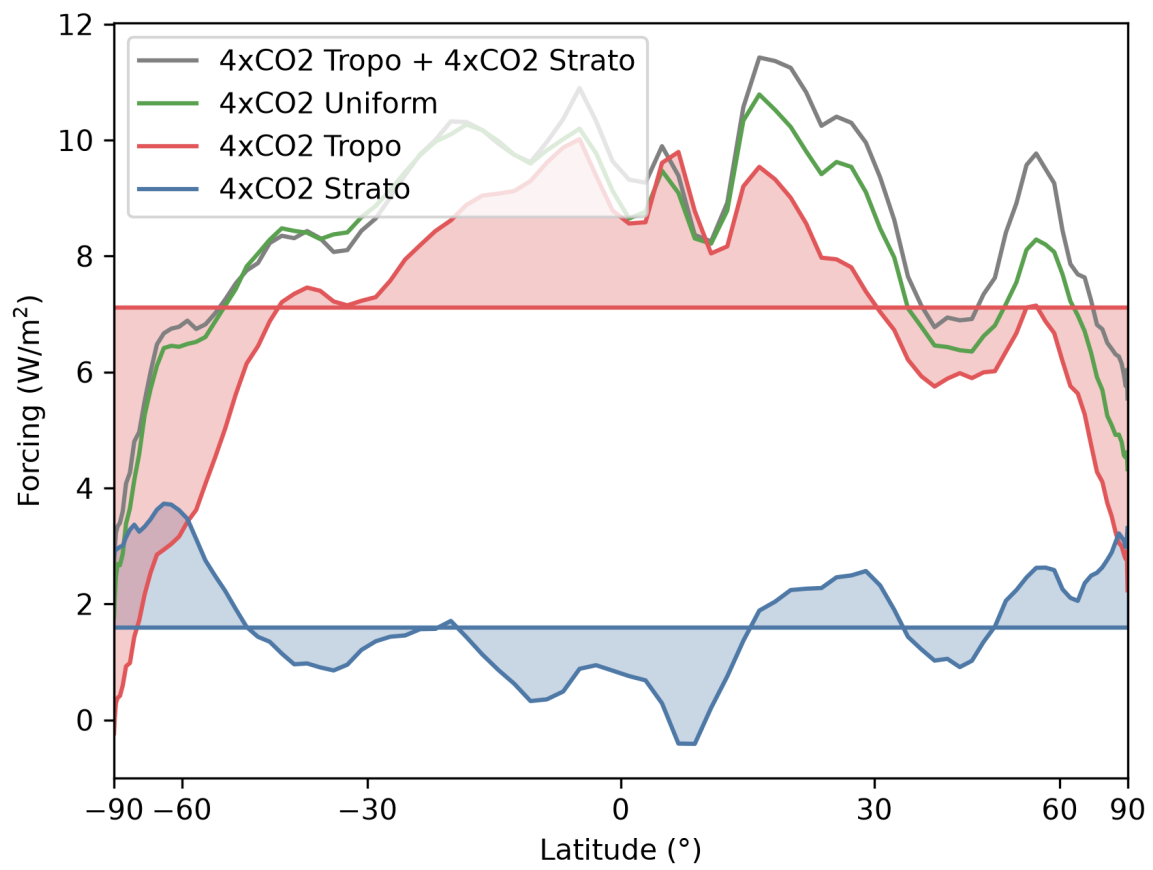


Figure 4.4

4.3. Eddy driven Hadley cell predictions

4.4. The energetic framework

4.4.1. GMS and energy transport of the Hadley cell

4.4.2. Eddy energy transport

4.4.3. Forcing and feedback contributions

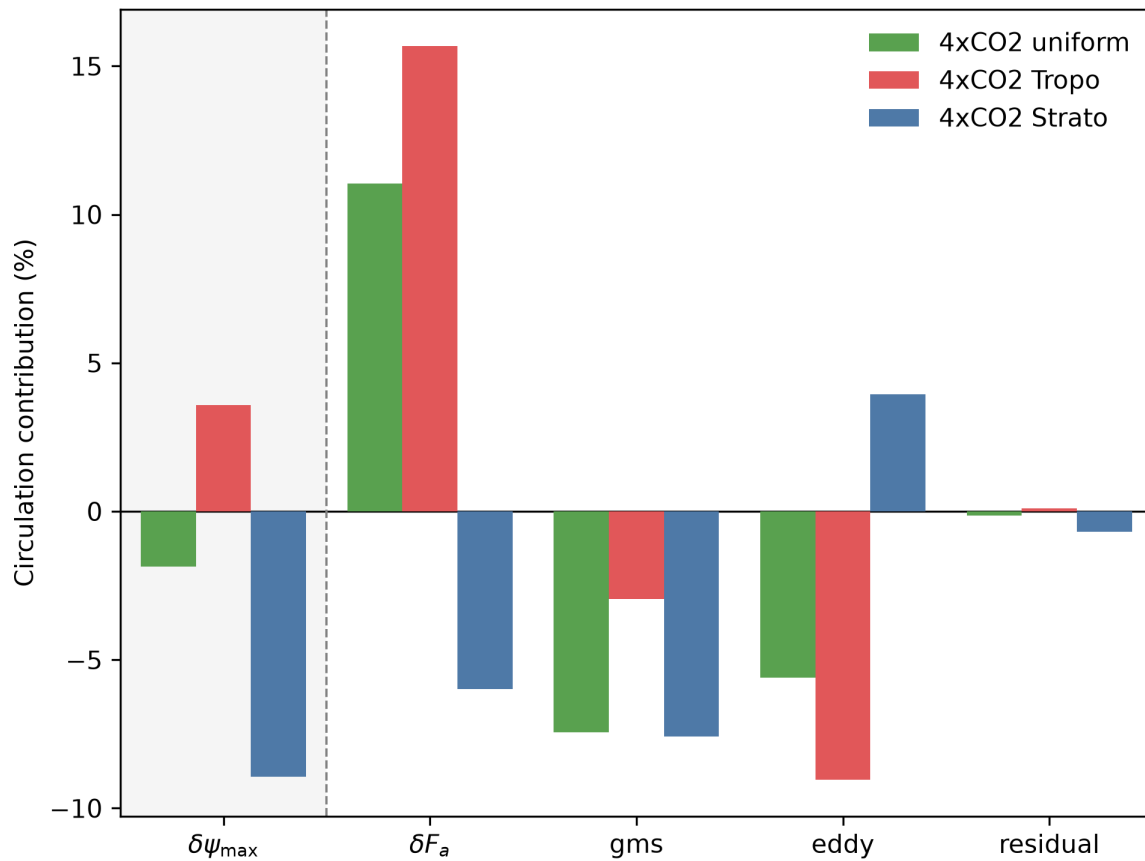


Figure 4.5

4.4.4. The forcing

Explain the forcing structure ([merlis_direct_2015](#))

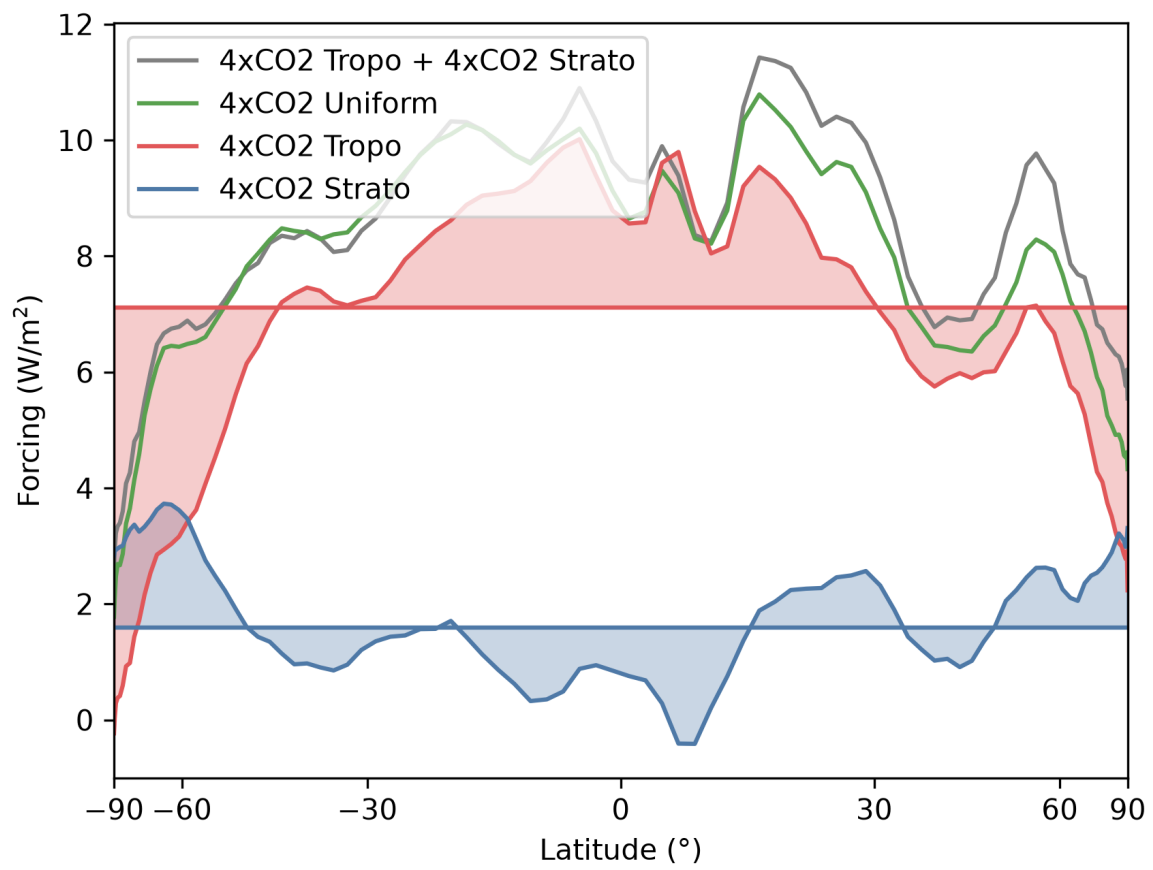


Figure 4.6

5

Discussion

6

Conclusion

Acknowledgement

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Thanks to for all the insights into Thanks to for all the intense but fruitful scientific debate about Thanks to ... , who always had an open door for

Also, I would like to thank the

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